Introduction to the Theory of Computation

Lecture 15. Gödel's First Incompleteness Theorem

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First-Order Peano Arithmetic (PA)

- **PA**: a formal theory of the **natural numbers** in **first-order logic with equality**.
- Language + axioms + inference rules.
- Gödel's theorem is about what PA can and cannot **prove**.

The Language of PA

$\mathcal{L}_{PA} = \{0, S, +, \times, =\}$

- 0: zero. S: successor ("$+1$").
- $+, \times$: addition, multiplication.
- $=$: equality.

Formulas: these symbols + variables + $\neg, \wedge, \vee, \rightarrow$ + $\forall, \exists$.

Origin: G. Peano, *Arithmetices principia, nova methodo exposita*, Turin: Bocca, 1889.

Logical Axioms and Rules

- PA includes all **axioms and inference rules of first-order logic**.
- E.g., quantifier axioms, *modus ponens*, universal generalization.

Details don't matter for us.

What Is a Formal Proof?

A **proof** of φ is a finite list of formulas

$$\varphi_1, \varphi_2, \ldots, \varphi_n = \varphi$$

where each φ_i is either

- an **axiom**, or
- obtained from earlier lines by an **inference rule**.

A proof is just a string. Checking it is a mechanical task.

Inference Rules: Only Two

Modus Ponens (MP).

$$\frac{\varphi \qquad \varphi \to \psi}{\psi}$$

Generalization (Gen).

$$\frac{\varphi}{\forall x\, \varphi}$$

Example.

- MP: from $0 = 0$ and $0 = 0 \to S(0) = S(0)$, get $S(0) = S(0)$.
- Gen: from $x + 0 = x$, get $\forall x\, (x + 0 = x)$.

Logical Axioms (1): Tautologies

Every tautology is an axiom.

Tautology = true under **every** true/false assignment.

- $p \vee \neg p$
- $p \to (q \to p)$

Replace p, q by any formulas, e.g.

$$(0 = S(0)) \vee \neg(0 = S(0))$$

Logical Axioms (2): Quantifiers and Equality

Instantiation. $\forall x\, \varphi(x) \;\to\; \varphi(t)$ for any term t

e.g. $\forall x\, (x + 0 = x) \;\to\; S(0) + 0 = S(0)$

Equality.

- $t = t$ (everything equals itself)
- $s = t \;\to\; \big(\varphi(s) \to \varphi(t)\big)$ (equals can be substituted)

e.g. with $\varphi(z) :\equiv z \times 0 = 0$: $x = y \;\to\; \big(x \times 0 = 0 \;\to\; y \times 0 = 0\big)$

(Plus one technical axiom for moving $\forall$. Shorthands: $\exists x \equiv \neg\forall x\neg$, $\wedge, \vee$ via $\neg, \to$.)

Arithmetic Axioms of PA

1. $\neg\big(S(x) = 0\big)$ — 0 is not a successor
2. $S(x) = S(y) \to x = y$ — S is injective
3. $x + 0 = x, \quad x + S(y) = S(x + y)$
4. $x \times 0 = 0, \quad x \times S(y) = x \times y + x$
5. **Induction schema** (next slides)

Example. Axioms 1, 2 give $1 \neq 2$: if $S(0) = S(S(0))$ then $0 = S(0)$, contradicting Axiom 1.

Axiom 5: Induction Schema

For **every** formula $\varphi(x)$:

$$\Big(\varphi(0) \ \wedge \ \forall x \left(\varphi(x) \to \varphi(S(x))\right)\Big) \ \to \ \forall x \, \varphi(x)$$

- Base case $+$ inductive step $\Rightarrow$ holds for all x.
- One axiom per formula: **infinitely many**, but easy to recognize.

Example: Proving $1 + 1 = 2$ in PA

Goal: $S(0) + S(0) = S(S(0))$.

1.	$\forall x \forall y \, (x + S(y) = S(x + y))$	Ax. 3, Gen
2.	$S(0) + S(0) = S\big(S(0) + 0\big)$	Inst, MP
3.	$\forall x \, (x + 0 = x)$	Ax. 3, Gen
4.	$S(0) + 0 = S(0)$	Inst, MP
5.	$(4) \to \big((2) \to \text{Goal}\big)$	Equality
6.	$S(0) + S(0) = S(S(0))$	MP twice

Tedious, but every step is **mechanically checkable**.

Why This Matters for Us

- Deciding whether a formula is an axiom: **algorithmic**.
- Checking whether a list is a valid proof: **algorithmic**.
- So a TM can **enumerate all proofs** of PA.

This is the only property of PA's proof system we will use.

Summary: What is PA?

$$\begin{aligned}\mathbf{PA} &= \text{first-order logic} \\ &+ \text{axioms for } 0, S, +, \times \\ &+ \text{induction schema}\end{aligned}$$

- Formalizes reasoning about $\mathbb{N}$.
- Proofs are **mechanically checkable**.

Expressing Numbers in PA

- No numerals like “5” built in — only 0 and S.

$$5 \equiv S(S(S(S(S(0)))))$$

- Shorthand: $\underline{n} \equiv \underbrace{S(S(\cdots S}_{n}(0)\cdots))$.

Examples.

$$2+2=4: \quad S(S(0))+S(S(0)) = S(S(S(S(0))))$$
$$2\times 3=6: \quad S(S(0))\times S(S(S(0))) = S(S(S(S(S(0))))))$$

Defining Relations with $\exists$

No $<$ or "divides" in the language — but we can **define** them:

$$
\begin{aligned}
x \leq y &:\equiv \exists z\,(x + z = y) \\
x < y &:\equiv x \leq y \wedge \neg(x = y) \\
x \mid y &:\equiv \exists k\,(x \times k = y) \\
\mathrm{Even}(x) &:\equiv \exists k\,(x = k + k)
\end{aligned}
$$

Example. $3 \mid 6$ unfolds to $\exists k\,(\underline{3} \times k = \underline{6})$, true with $k = \underline{2}$.

Defining Primes with $\forall$

$$\mathrm{Prime}(p) :\equiv \underline{1} < p \wedge \forall a \, \forall b \, \left(p = a \times b \rightarrow a = \underline{1} \vee b = \underline{1}\right)$$

"$p > 1$, and every factorization of p is trivial."

Examples.

- $\mathrm{Prime}(\underline{5})$: true — only $5 = 1 \times 5 = 5 \times 1$.
- $\mathrm{Prime}(\underline{6})$: false — $a = \underline{2}$, $b = \underline{3}$ breaks the $\forall$.

Known Theorems in PA (1)

Every number is even or odd.

$$\forall x \, \exists k \, \big(x = k + k \ \vee \ x = S(k + k) \big)$$

Infinitely many primes (Euclid).

$$\forall n \, \exists p \, \big(n < p \ \wedge \ \mathrm{Prime}(p) \big)$$

Known Theorems in PA (2)

$\sqrt{2}$ **is irrational.**

$$\neg\exists a\,\exists b\,\big(\neg(b = 0)\ \wedge\ a \times a = \underline{2} \times b \times b\big)$$

Four-square theorem (Lagrange, 1770).

$$\forall n\,\exists a\,\exists b\,\exists c\,\exists d\,\big(n\ = a \times a + b \times b + c \times c + d \times d\big)$$

Famous Open Conjectures in PA

Goldbach's conjecture (1742). Every even $n > 2$ is a sum of two primes.

$$\forall n \left(\underline{2} < n \wedge \mathrm{Even}(n) \rightarrow \exists p \, \exists q \left(\mathrm{Prime}(p) \wedge \mathrm{Prime}(q) \wedge n = p + q \right) \right)$$

Twin prime conjecture. Infinitely many p with p, $p + 2$ both prime.

$$\forall n \, \exists p \left(n < p \wedge \mathrm{Prime}(p) \wedge \mathrm{Prime}(p + \underline{2}) \right)$$

In-Class Exercise

Write each as a PA formula.

1. x is **odd**.
2. y is a **perfect square**.
3. y is a **sum of two squares**.
4. z is the **maximum** of x and y.
5. x and y are **coprime**.

For 4–5, you may use $x \leq y$ and $x \mid y$.

From PA to Lean

- **Lean**: a formal proof system, like PA, but far more expressive.
- You can add your own axioms.
- Given the axioms, checking a proof is **mechanical** — done by a program.

Gödel's First Incompleteness Theorem

Theorem 1 (Gödel, 1931)
Let F be a ***consistent*** *formal system that contains PA and has* ***mechanically checkable*** *proofs. Then there is a formula ϕ such that* ***neither*** ϕ ***nor*** $\neg\phi$ *is provable in F.*



Kurt Gödel, 1925

Proof Strategy: Reduction from PCP

- Suppose F is **complete**: every ϕ is provable or disprovable.
- Then we can use F's **proof system** to **decide** L_{pcp}: search all proofs until one of $\phi(P)$ or $\neg\phi(P)$ is found.
- But L_{pcp} is **undecidable**. Contradiction!

The Decision Procedure (If F Were Complete)

Suppose for every PCP instance P we can build a PA formula $\phi(P)$ with

$$P \text{ has a match} \iff \phi(P) \text{ is true.}$$

Then this TM decides L_{pcp}:

1. Given P, construct $\phi(P)$.
2. For $\ell = 1, 2, \ldots$: check every proof of length ℓ. Does it prove $\phi(P)$ or $\neg\phi(P)$?
3. Found one: output true/false accordingly and halt.

From Strings to Numbers: An Example

Strings $\to$ numbers: read 1s in binary. The leading 1 keeps leading zeros.

$$P = \left\{ \left[\frac{1}{111}\right], \left[\frac{10111}{10}\right], \left[\frac{10}{0}\right] \right\}, \qquad \text{match: } 2, 1, 1, 3.$$

i	top prefix x_i	bottom prefix y_i
0	1	1
1	1 10111	1 10
2	1 10111 1	1 10 111
3	1 10111 1 1	1 10 111 111
4	1 10111 1 1 10	1 10 111 111 0

Each step: $x_i = x_{i-1} \cdot 2^{|t_k|} + (t_k)_2$. Match $\iff x_m = y_m$.

The Formula $\phi(P)$

Let $P = \{[t_1/b_1], \ldots, [t_n/b_n]\}$.

$\phi(P)$ says: there exist $m \geq 1$ and sequences $x_0, \ldots, x_m$, $y_0, \ldots, y_m$ with

$$x_0 = y_0 = 1, \qquad x_m = y_m,$$

$$(\forall i \in [1, m]) \bigvee_{k=1}^{n} \begin{pmatrix} x_i = x_{i-1} \cdot 2^{|t_k|} + (t_k)_2 \\ \wedge \; y_i = y_{i-1} \cdot 2^{|b_k|} + (b_k)_2 \end{pmatrix}.$$

$2^{|t_k|}$, $(t_k)_2$, ... are **constants**.

Only problem: PA cannot say "there exists a sequence".

$\phi(P)$ for Our Example, Written Out

$$P = \left\{ \left[\tfrac{1}{111}\right], \left[\tfrac{10111}{10}\right], \left[\tfrac{10}{0}\right] \right\}.$$

$$\exists m \geq 1 \;\; \exists X_0, \ldots, X_m \;\; \exists Y_0, \ldots, Y_m \left[X_0 = Y_0 = 1 \;\wedge\; X_m = Y_m \right.$$

$$\wedge \;\; (\forall i \in [1, m]) \Big($$

$$\big(X_i = 2X_{i-1} + 1 \;\wedge\; Y_i = 8Y_{i-1} + 7\big) \quad \left[\tfrac{1}{111}\right]$$

$$\vee \;\; \big(X_i = 32X_{i-1} + 23 \;\wedge\; Y_i = 4Y_{i-1} + 2\big) \quad \left[\tfrac{10111}{10}\right]$$

$$\left. \vee \;\; \big(X_i = 4X_{i-1} + 2 \;\wedge\; Y_i = 2Y_{i-1}\big) \Big) \right] \quad \left[\tfrac{10}{0}\right]$$

Still to fix: "$\exists X_0, \ldots, X_m$" — the number of variables depends on m.

Gödel's Beta Function: Sequences as Numbers

$$\beta(a, b, i) = a \bmod \big(1 + (i+1)\, b\big)$$

is expressible in PA: $\beta(a, b, i) = r \iff \exists q \big(a = q(1 + (i+1)b) + r \wedge r < 1 + (i+1)b\big)$.

Lemma 2
For any $x_0, \ldots, x_m$, there exist a, b with $\beta(a, b, i) = x_i$ for all $i \leq m$.

So "$\exists$ sequence x" becomes "$\exists a\, \exists b$", with $x_i := \beta(a, b, i)$.

Proof of the Lemma: Chinese Remainder Theorem

CRT: If $m_0, \ldots, m_k$ are pairwise coprime, then for any $r_i < m_i$ there is a with $a \bmod m_i = r_i$ for all i.

Let $M = \max(m, x_0, \ldots, x_m)$, $b = M!$, $m_i = 1 + (i+1)\, b$.

- $m_i > b \geq M \geq x_i$: remainders fit.
- Pairwise coprime: if prime $p \mid m_i, m_j$ $(i < j)$, then $p \mid (j - i)\, b$.
  - $p \nmid b$, since $m_i \equiv 1 \pmod{p}$ otherwise.
  - So $p \mid j - i \leq m \leq M$, hence $p \mid M! = b$. Contradiction.

CRT gives a. □

$\phi(P)$ with β Plugged In

$$
\begin{aligned}
&\exists m\, \exists a\, \exists b\, \exists c\, \exists d\, \Big[\ m \geq 1 \\
&\quad \wedge\ a \bmod (1+b) = 1 \ \wedge\ c \bmod (1+d) = 1 \\
&\quad \wedge\ a \bmod \big(1+(m+1)b\big) = c \bmod \big(1+(m+1)d\big) \\
&\quad \wedge\ (\forall j < m)\ \exists u\, \exists u'\, \exists v\, \exists v'\, \Big(\\
&\qquad u = a \bmod \big(1+(j+1)b\big)\ \wedge\ u' = a \bmod \big(1+(j+2)b\big) \\
&\qquad \wedge\ v = c \bmod \big(1+(j+1)d\big)\ \wedge\ v' = c \bmod \big(1+(j+2)d\big) \\
&\qquad \wedge\ \big[\ (u' = 2u+1\ \wedge\ v' = 8v+7) \quad \big[\tfrac{1}{111}\big] \\
&\qquad\quad \vee\ (u' = 32u+23\ \wedge\ v' = 4v+2) \quad \big[\tfrac{10111}{10}\big] \\
&\qquad\quad \vee\ (u' = 4u+2\ \wedge\ v' = 2v)\ \big]\ \Big)\Big] \quad \big[\tfrac{10}{0}\big]
\end{aligned}
$$

$$x = a \bmod n \ \text{ means } \ \exists q\,(a = qn + x\ \wedge\ x < n).$$

Putting It Together: Contradiction

- We built a PA formula $\phi(P)$ with P has a match $\iff$ $\phi(P)$ is true.
- If F is complete, the decision procedure **decides** L_{pcp}.
- But L_{pcp} is **undecidable**. **Contradiction.**
- Hence some ϕ has neither ϕ nor $\neg\phi$ provable in F. ■

Is Incompleteness Good News?

- Hilbert's dream: mathematics = mechanical symbol manipulation.
- Gödel (1931), Turing (1936): **this is impossible**.
- No fixed formal system captures all mathematical truth.

Mathematics needs insight, not just syntax.

Minds vs. Machines?

Can machines (AI) replace humans? If not, what capabilities are **uniquely human**?